

B.Tech. Degree I Semester Regular Examination November 2023

EE/EC/CS/IT 23-200-0105B COMPUTER PROGRAMMING
(2023 Scheme)

Maximum Marks: 50

Time: 3 Hours

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Identify main components of a computer system and explain its working.
 CO2: Develop flow chart and algorithms for computational problems.
 CO3: Write the syntax of various constructs of C language.
 CO4: Build efficient programs by choosing appropriate decision-making statements, loops and data structures.
 CO5: Design modular programs using functions for larger problems.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create
 PO – Programme Outcome

PART A

(Answer **ALL** questions)

(5 × 2 = 10)

	Marks	BL	CO	PO
I. (a) Explain the various categories of software.	2	L1	1	1,2,3
(b) Is it advisable to use goto statements in a C program? Justify your answer.	2	L2	3	1,2,3
(c) What is the output of following program? Explain the answer.	2	L3	4	1,2,3
<pre>#include <stdio.h> int main () { int arr [] = {1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip = arr+4; printf("%d\n", ip[1]); return 0; }</pre>				
(d) Explain dynamic memory allocation in C.	2	L2	3	1,2,3
(e) What is the output of following program? Explain the answer.	2	L3	4	1,2,3
<pre>#include <stdio.h> int main() { enum days{ mon=-1,tues,we=4,thu,fri,sat}; Printf("%d,%d,%d,%d,%d,%d",mon,tue,wed,thu,fri,sat); return 0; }</pre>				

PART B

(4 × 10 = 40)

- II. (a) Explain how linker, loader and interpreter contribute to the overall process of program execution? 5 L2 1 1,2,3
- (b) Explain the concept of variables and how data types are associated with them. In the following programming scenarios recommend an appropriate data type for a variable based on nature of data it will store. 5 L3 2 1,2,3
- (i) Mark
- (ii) Temperature
- (iii) Employee name
- (iv) Student details

OR

(P.T.O.)

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		Marks	BL	CO	PO
III.	(a) Write an algorithm and draw flowchart for the given sequence: Input purchase amount, if purchase amount greater than 10000 then give discount of 5% else 2%, calculate and display discount and net amount.	5	L6	2	1,2,3
	(b) What are the primary components of a digital computer system and how do they interact to perform computations?	5	L6	1	1,2,3
IV.	(a) Provide a detailed explanation of entry-controlled and exit-controlled loops, along with appropriate example.	5	L2	3	1,2,3
	(b) Write a program to display the following output 5 4 4 3 3 3 2 2 2 2 1 1 1 1 1	5	L6	4	1,2,3
OR					
V.	(a) Design and develop a C program to reverse an integer number NUM and check whether it is PALINDROME or NOT.	6	L6	4	1,2,3
	(b) Is there any difference between the pre-decrement and post decrement operators? Explain with suitable examples.	4	L3	3	1,2,3
VI.	(a) Write a program that reads a sentence and print the frequency of each vowels and total count of consonants.	6	L6	4	1,2,3
	(b) What is recursion? Write a program in C to find out factorial of a given number using recursion.	4	L2	5	1,2,3
OR					
VII.	(a) Write a C program that implements string copy operation that copies a string str1 to another string str2 without using library function.	4	L6	4	1,2,3
	(b) Explain the concept of function parameter and their role in passing data to a function. Discuss the difference between call by value and call by reference.	6	L2	5	1,2,3
VIII.	(a) Write a program in C to sort an array using a pointer.	7	L6	5	1,2,3
	(b) Illustrate difference between structure and union.	3	L2	3	1,2,3
OR					
IX.	(a) Write a short note on pointer arithmetic.	5	L1	3	2,3
	(b) Write a program that allows the user to input details for multiple employees including their salary, and calculate the average salary (using structure).	5	L6	5	2,3

Bloom's Taxonomy Levels

L1 = 10%, L2 = 38%, L3 = 19%, L6 = 33%.
